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Question Bank: MAT 101. Linear Algebra and Calculus

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Total Number of Questions 264

Module 1

1) Solve the System

$$2x_1 + x_2 + 2x_3 + x_4 = 6$$

$$6x_1 - 6x_2 + 6x_3 + 12x_4 = 36$$

$$4x_1 + 3x_2 + 3x_3 - 3x_4 = -1$$

$$2x_1 + 2x_2 - x_3 + x_4 = 10$$

7 Mark

2. Solve the System of equation

$$x + 2y - 3z = 0 \quad (1)$$

$$2x - 3y + z = 0 \quad (2)$$

$$4x - y - 2z = 0 \quad (3) \quad 7 \text{ Mark}$$

3. Solve the System

$$x_1 + 2x_2 - x_3 + 2x_4 = 0$$

$$2x_1 + 4x_2 + 14x_4 = 0$$

$$-x_1 - 2x_2 + 4x_3 + 13x_4 = 0 \quad 7 \text{ Mark}$$

4. Solve the System of equation

$$x + 2y - 3z = 0 \quad (1)$$

$$2x - 3y + z = 0 \quad (2)$$

$$4x - y - 2z = 0 \quad (3) \quad 7 \text{ Mark}$$

5. Solve the System

$$x_1 + 2x_2 - x_3 + 2x_4 = 0$$

$$2x_1 + 4x_2 + 14x_4 = 0$$

$$-x_1 - 2x_2 + 4x_3 + 13x_4 = 0 \quad 7 \text{ Mark}$$

6: Find the rank and a basis for row space and basis for

Column space of the matrix

$$\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & -4 \\ 0 & 4 & 0 \end{bmatrix} \quad 7 \text{ Mark}$$

7 : Find a basis for row space ,column space and null space of the matrix

$$A = \begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ -2 & -5 & 1 & -1 & -8 \\ 0 & -3 & 3 & 4 & 1 \\ 3 & 6 & 0 & -7 & 2 \end{bmatrix} \quad 7 \text{ Mark}$$

8. : Prove that the vectors $(1, -1, 1)$, $(0, 1, 2)$ and $(3, 0, -1)$ are linearly independent ,or this vectors form a basis for R^3

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 2 \\ 3 & 0 & -1 \end{bmatrix} \quad 7 \text{ Mark}$$

9 : Prove that the vectors $(2, -1, 3, 2)$, $(1, 3, 4, 2)$ and $(3, -5, 2, 2)$ are linearly dependent in R^4 7 Mark

10: Find the eigen values and eigenvectors of the matrix

$$A = \begin{bmatrix} 8 & -4 \\ 2 & 2 \end{bmatrix} \quad 7 \text{ Mark}$$

11: Find the eigen values and eigen vectors of the matrix

$$A = \begin{bmatrix} 7 & -1 \\ 4 & 3 \end{bmatrix} \quad 7 \text{ Mark}$$

12 Find the eigen values and eigen vectors of the matrix

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix} \quad 7 \text{ Mark}$$

13: Find the eigen values and eigen vectors of the matrix

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix} \quad 7 \text{ Mark}$$

14: Diagonalise the matrix $A = \begin{bmatrix} 8 & -4 \\ 2 & 2 \end{bmatrix}$ 7 Mark

15: Diagonalise the matrix $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$ 7 Mark

16: Find the eigen values and eigen vectors of the matrix

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix} \quad 7 \text{ Mark}$$

17: Examine whether the matrix $A = \begin{bmatrix} 7 & -1 \\ 4 & 3 \end{bmatrix}$ diagonalizable. 7 Mark

18: Diagonalize the real symmetric matrix $A = \begin{bmatrix} 3 & -2 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ 7 Mark

19 Diagonalize the real symmetric matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ 7 Mark

20 : Find the principal axes form or Canonical form or diagonal form of the quadratic form $q(X) = 2x_1^2 + x_2^2 + x_3^2 + 2x_1x_2 - 2x_1x_3 - 4x_2x_3$. Also find the orthogonal transformation $X = PY$ ie $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = PY$ or express $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ in terms of new coordinate $\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$.

Where P is the model matrix of the symmetric matrix associate to the quadratic form.

$$q(X) = 2x_1^2 + x_2^2 + x_3^2 + 2x_1x_2 - 2x_1x_3 - 4x_2x_3 \quad 7 \text{ Mark}$$

21: Find the canonical form of the quadratic $3x^2+2xy+3y^2$. Hence show that the equation $3x^2+2xy+3y^2-8=0$ represents an ellipse in R^2 . Also find X in terms of new co-ordinate.
7 Mark

22: Find the nature, rank and signature of the quadratic form

$$3x^2+5y^2+3z^2-2xy+2xz-2yz \quad 3 \text{ Mark}$$

23: Find the nature, rank and signature of the quadratic form

$$x^2+2y^2+z^2-2xy+2xz \quad 3 \text{ Mark}$$

24: What kind of conic section or pair of straight line is given by the

$$\text{quadratic form } 3x^2+22xy+3y^2=0 \quad 7 \text{ Mark}$$

25: What kind of conic section or pair of straight line is given by the

$$\text{quadratic form } x^2-12xy+y^2=70 \quad 3 \text{ Mark}$$

Module 2

1 Find $\lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 - 4xy}{\sqrt{x} - 2\sqrt{y}}.$ 3 Mark

2 Consider the function $f(x, y) = \frac{4x^6 y^2}{x^{12} + y^4}.$ 6*3 Mark

Find the limit of $f(x, y)$ as $(x, y) \rightarrow (0, 0)$ along

(a) the x -axis

(b) the y -axis

(c) the line $y=x$

(d) the line $y=-x$

(e) the parabola $y=x^2$

(f) the curve $y=x^3$

3 Using two path test, show that

$$f(x, y) = \frac{4x^6 y^2}{x^{12} + y^4}$$

has no limit as (x, y) approaches $(0, 0)$. 3 Mark

4. Show that

$$f(x, y) = \begin{cases} \frac{4x^2 y}{x^3 + y^3}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases} \quad 7 \text{ Mark}$$

is continuous at every point except the origin.

5. Can the function $(x^2 + y^2) \ln(x^2 + y^2)$ be continuous at

$$(0, 0). \quad \text{Find } \lim_{(x, y) \rightarrow (0, 0)} (x^2 + y^2) \ln(x^2 + y^2).$$

7 Mark

6. Evaluate the limit **Example 16** If $f(x, y) = x + y$,

$$\text{find } \frac{\partial f}{\partial x} \text{ and } \frac{\partial f}{\partial y}. \quad 3 \text{ Mark}$$

7. Find the value of f_x at the point $\left(2, \frac{\pi}{4}\right)$ if

$$f(x, y) = x \sin xy. \quad 3 \text{ Mark}$$

8. If $f(x, y) = y \exp x$, find the first order partial derivatives of f .

3 Mark

9. Find the first order partial derivatives at the point $(2, 0)$
of the function f when

$$f(x, y) = 8x^3y + 4xy^2 - 12x + 8y - 10. \quad 3 \text{ Mark}$$

10. Find the first order partial derivatives of f when

$$f(x, y) = x \tan^{-1}(xy) \quad 3 \text{ Mark}$$

11. If $f(x, y) = \frac{2y}{y + \cos x}$, find f_x . 3 Mark

12. Let $f(x, y) = x^2y + 5y^3$.

(a) Find the slope of the surface $z = f(x, y)$ in the x -direction at the point $(1, -2)$. 3 Mark

(b) Find the slope of the surface $z = f(x, y)$ in the y -direction at the point $(1, -2)$. 3 Mark

13. Discuss the partial derivatives of the function

$$f(x, y) = \begin{cases} -\frac{xy}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0) \end{cases} \quad 7 \text{ Mark}$$

14. The plane $x = 2$ intersects the paraboloid $z = x^2 + y^2$ in a parabola.

Find the slope of the tangent to the parabola at $(2, 3, 13)$, 3 Mark

15. Find $\frac{\partial z}{\partial x}$ if the equation $yz - \ln z = x + y$ defines z as a function of the two independent variables x and y . 3 Mark

16. Suppose that $D = \sqrt{x^2 + y^2}$ is the length of the diagonal of a rectangle whose sides has lengths x and y that are allowed to vary. Find a formula for the rate of change of D with respect to x if x varies with y held constant, and use this formula to find the rate of change of D with respect to x at the

point where $x = 3$ and $y = 4$. 7 Mark

17. Find the slope of the sphere $x^2 + y^2 + z^2 = 1$ in the y -direction at the points $(\frac{2}{3}, \frac{1}{3}, \frac{2}{3})$ and $(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3})$. 7 Mark

18. Compute all the first and second partial derivatives of the function

$$f(x, y) = x^2 y \quad . \quad 7 \text{ Mark}$$

19. Verify that $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$, where $f = x^y$. 3 Mark

20. Find $\frac{\partial^2 f}{\partial x \partial y}$, where $f(x, y) = xy + \frac{e^y}{y^2 + 1}$. 7 Mark

21. Verify $\frac{\partial^3 u}{\partial y \partial x^2} = \frac{\partial^3 u}{\partial x^2 \partial y}$, where $u = y^2 e^x + x^2 y^3 + 16$. 3 Mark

22. If $z = e^x (x \cos y - y \sin y)$,

prove that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$. 3 Mark

23. If $z = \tan^{-1}\left(\frac{y}{x}\right)$, prove that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$.

7 Mark

24. Find all first and second order partial derivatives of the function

$$f(x, y) = x \cos y + ye^x.$$

7 Mark

25. If x , y , and z are independent variables and

$$w = f(x, y, z) = x \sin(y + 3z), \text{ find } \frac{\partial w}{\partial z}, \frac{\partial^2 w}{\partial x \partial z}$$

$$\text{and } \frac{\partial^3 w}{\partial x^2 \partial z}.$$

7 Mark

26. If $u = \log(\tan x + \tan y + \tan z)$, then prove that

$$\sum \sin 2x \cdot \frac{\partial u}{\partial x} = 2.$$

7 Mark

27. If $v = f\left(\frac{x}{z}, \frac{y}{z}\right)$, show that $x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} + z \frac{\partial v}{\partial z} = 0$.

7 Mark

28. If $v = \frac{1}{(x^2 + y^2 + z^2)^{1/2}}$, with $x^2 + y^2 + z^2 \neq 0$

,

show that $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0.$ 7 Mark

29. Show that the function

$u(x, t) = \sin(x - ct)$ is a solution of the
one dimensional wave equation. 3 Mark

30. Verify that $u = x^3 + 3xt^2$ is a solution of the one dimensional
wave equation (8), for a suitable value of c . 3 Mark

31. Show that the functions

$u = x^2 - y^2,$ $u = e^x \cos y,$
 $u = \log(x^2 + y^2)$

are *solutions* of the two dimensional Laplace equation

$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. 7 Mark

32. Let $f(x, y) = x^2 + y^2$. Prove that f is differentiable at
(0, 0).

7 Mark

33. Estimate the change in $z = xy^2$ from its value at (0.5, 1.0) to its value at
(0.503, 1.004). Compare the magnitude of the error in this approximation
with the distance between the points (0.5, 1.0) to its value at (0.503, 1.004).

7 Mark

34. Estimate the resulting absolute, relative, and percentage changes in the values

of the function $f = f(x, y) = \pi x^2 y$, when the variables x and y change from the initial values of $(x_0, y_0) = (1, 5)$ by the amounts

$$dx = 0.03 \quad \text{and} \quad dy = -0.1. \quad 7 \text{ Mark}$$

35. The volume $V = \pi r^2 h$ of a right circular cylinder is to be calculated from measured values of r and h . Suppose that r is measured with an error of no more than **2%** and h with an error of no more than **0.5%**. Estimate the resulting possible percentage error in the calculation of V . 7 Mark
36. The length, width, and height of a rectangular box are measured with an error of at most **5%**. Use a total differential to estimate the maximum percentage error that results if these quantities are used to calculate the diagonal of the box. 7 Mark

37. Let $L(x, y)$ denote the local linear approximation to

$$f(x, y) = \sqrt{x^2 + y^2} \quad \text{at the point } (3, 4). \quad \text{Compare the error in}$$

$$\text{approximating } f(2.98, 4.01) = \sqrt{(2.98)^2 + (4.01)^2}$$

$$\text{by } L(2.98, 4.01) \quad \text{with the distance between the points } (3, 4)$$

$$\text{and } (2.98, 4.01). \quad 7 \text{ Mark}$$

38. Find the linearization $L(x, y, z)$ of

$$f(x, y, z) = x^2 - xy + 3 \sin z$$

$$\text{at the point } (x_0, y_0, z_0) = (2, 1, 0). \quad 7 \text{ Mark}$$

39. Find $\frac{dy}{dt}$, when $y = \sin x$, where $x = t^2$. 7 Mark

40. Find $\frac{dw}{dt}$ using chain rule, when

$$w = xy \text{ with } x = \cos t \text{ and } y = \sin t.$$

$$t = \frac{\pi}{4}$$

What is the derivative's value at 7 Mark

41. Use the chain rule to find the derivative of $w = xy$ with respect to θ along the path $x = \cos \theta$, $y = \sin \theta$. What is the derivative's value at

$$\theta = \frac{\pi}{2}?$$

7 Mark

42. Find $\frac{dw}{dt}$ if $w = xy + z$, $x = \cos t$, $y = \sin t$, $z = t$.

What is the derivative's value at $t = \frac{\pi}{2}$. 7 Mark

43. By implicit differentiation, find $\frac{dy}{dx}$ if

$$x^2 + \sin y - 2y = 0. \quad 7 \text{ Mark}$$

44. At what rate is the volume of a box changing if its length is 8 ft and increasing at 3 ft/s, its width is 6 ft and increasing at 2 ft/s, and its height is 4 ft and increasing at 1 ft/s? 7 Mark

45. Express $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$ in terms of u and v if

$$w = x + 2y + z^2, \quad x = \frac{u}{v}, \quad y = u^2 + \ln v, \quad z = 2u.$$

7 Mark

46. Express $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$ in terms of r and s if

$$w = x^2 + y^2, \quad x = r - s, \quad y = r + s. \quad 7 \text{ Mark}$$

47. If w is a function of x and y and

$$x = e^u + e^{-v}, y = e^{-u} - e^v$$

then prove that $\frac{\partial w}{\partial u} - \frac{\partial w}{\partial v} = x \frac{\partial w}{\partial x} - y \frac{\partial w}{\partial y}.$ 7 Mark

48. If $v = \frac{1}{r}$ and $r^2 = x^2 + y^2$, show that $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{1}{r^3}.$

7 Mark

49. If $v = f\left(\frac{x}{z}, \frac{y}{z}\right)$, show that $x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} + z \frac{\partial v}{\partial z} = 0.$ 7 Mark

50. Find the total derivative of $f(x, y) = \pi x^2 y$ with respect to x ,

given that $y = \cos x.$ 7 Mark

51. Find the absolute maximum and minimum values of

$$f(x, y) = 2 + 2x + 2y - x^2 - y^2$$

on the triangular plate in the quadrant bounded by the lines

$$x = 0, y = 9 - x.$$

7 Mark

52. Discuss the relative extrema and absolute extrema of the function

$$f(x, y) = x^2 + y^2.$$

7 Mark

53. Discuss the relative extrema of the function

$$f(x, y) = 1 - x^2 - y^2.$$

7 Mark

54. Discuss the relative extrema of the function

$$f(x, y) = \sqrt{x^2 + y^2}.$$

7 Mark

55. By means of an example, show that a function of two variables need not have a relative extremum at every stationary point.

7 Mark

56. Find the local extreme values (if any) of $f(x, y) = y^2 - x^2$.

7 Mark

57. Find the relative extreme values of the function

$$f(x, y) = xy - x^2 - y^2 - 2x - 2y + 4.$$

7 Mark

58. Locate all relative extrema and saddle points of

$$f(x, y) = xy.$$

7 Mark

59. Show that the function $f(x, y) = x^3 \exp(-x^2 - y^2)$ has a maximum at the point $(\sqrt{3}/2, 0)$, a minimum at $(-\sqrt{3}/2, 0)$ and a stationary point at the origin whose nature cannot be determined by the above procedures.

7 Mark

Module 3

1. Evaluate the double integral

$$\iint_R y^2 x \, dA$$

over the rectangle $R = \{(x, y) : -3 \leq x \leq 2, 0 \leq y \leq 1\}$.

3 Mark

2. Evaluate $\iint_R f(x, y) \, dA$ for

$$f(x, y) = 1 - 6x^2 y, \quad R : 0 \leq x \leq 2, -1 \leq y \leq 1.$$

Verify that the change in the order of integration doesn't effect the result.

3 Mark

3. Evaluate $\int_0^1 \int_0^2 xy(x - y) \, dx \, dy$

3 Mark

4. Evaluate $I = \int_0^1 \int_1^2 (x^2 + y^2) \, dx \, dy.$

3 Mark

5. Evaluate $I = \int_0^3 \int_0^2 xy \, dy \, dx$

3 Mark

6. Use a double integral to find the volume of the solid that is bounded above by the plane

$$z = 4 - x - y \quad \text{and below by the rectangle}$$

$$R = [0, 1] \times [0, 2].$$

3 Mark

$$I = \int_2^a \int_2^b \frac{1}{xy} dx dy$$

7. Evaluate

3 Mark

$$\int_0^1 \int_x^2 x \, dx \, dy$$

8. Evaluate

3 Mark

$$\int_0^a \int_0^x \sqrt{x^2 + y^2} \, dy \, dx$$

9. Evaluate

3 Mark

$$\int_1^2 \int_0^x \frac{dx \, dy}{x^2 + y^2}$$

10. Evaluate

3 Mark

11. Evaluate $\iint r \sqrt{a^2 - r^2} \, dr \, d\theta$ over the upper

half of the circle $r = a \cos \theta$. 3 Mark

12. Evaluate

$$\iint_R xy \, dA$$

over the region R enclosed between $y = \frac{1}{2}x$,

$$y = \sqrt{x}, \, x = 2, \, \text{and} \, x = 4.$$

7 Mark

13. Evaluate $\iint_R xy \, dx \, dy$ over the region R where R is the first quadrant of the circle $x^2 + y^2 = a^2$. 7 Mark

14. Evaluate $\iint_S (x^2 + y^2) \, dx \, dy$ over the region S in which $x \geq 0$; $y \geq 0$ and $x + y \leq 1$. 3 Mark

15. Evaluate $\iint_D (x + y) \, dx \, dy$ where D is the domain in the first quadrant of the circle

$$x^2 + y^2 = 9. \quad 7 \text{ Mark}$$

16. Use a double integral to find the volume of the tetrahedron bounded by the coordinate planes and the plane $z = 4 - 4x - 2y$. 3 Mark

17. Find the volume of the solid bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4$ and $z = 0$. 7 Mark

18. Evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} \, dy \, dx$ by changing the order of integration. 7 Mark

19. Change the order of integration and hence evaluate the double integral

$$\int_0^1 \int_{e^x}^e \frac{dx \, dy}{\log y}$$

3 Mark

20. Change the order of integration and hence evaluate the double integral

$$\int_0^1 \int_x^{2-x} \frac{x}{y} \, dy \, dx$$

3 Mark

21. Evaluate $\int_0^2 \int_{y/2}^1 e^{x^2} dx dy$

3 Mark

22. Calculate $\iint_R \frac{\sin x}{x} dA$, where R is the triangle in the xy -plane bounded

by the x -axis, the line $y = x$, and the line $x = 1$. Whether changing the order of integration works. 7 Mark

23. Evaluate $\iint_R \sin \theta dA$ where R is the region that is outside the circle

$r = 1$ and inside the cardioid $r = 1 + \cos \theta$. 7 Mark

24. Evaluate $\iint_R \sin \theta dA$ where R is the region in the first quadrant that is

outside the circle $r = 2$ and inside the cardioid $r = 2(1 + \cos \theta)$.

7 Mark

25. The sphere of radius a centered at the origin is expressed in rectangular coordinates as $x^2 + y^2 + z^2 = a^2$, and hence its equation in cylindrical

coordinates is $r^2 + z^2 = a^2$. Use this equation and a polar double integral to find the volume of the sphere. 7 Mark

26. Find the area enclosed by the lemniscates $r^2 = 4 \cos 2\theta$. 3 Mark

27. Find the area enclosed by the cardioid $r = a(1 + \cos \theta)$. 7 Mark

28. Use a polar double integral to find the area enclosed by the three-petaled rose $r = \sin 3\theta$. 7 Mark

29. Evaluate

$$\iint_R e^{x^2+y^2} dy dx,$$

where R is the semicircular region bounded by

the x-axis and the curve $y = \sqrt{1-x^2}$ 7 Mark

30. Use polar coordinates to evaluate

$$\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)^{3/2} dy dx. \quad 7 \text{ Mark}$$

31. Find the area enclosed between $x = 5$, $x = 10$ and $y = x$ and $y = 5 + x$. 7 Mark

32. Find the area bounded between the curve $y = x^2$ above the x-axis and below the line $y = 2$ 7 Mark

33. Find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.7 Mark

34. Use a double integral to find the area of the region R enclosed between the

parabola $y = \frac{1}{2}x^2$ and the line $y = 2x$. 7 Mark

35. Find the volume enclosed by the co-ordinate planes and the portion of the plane $x + y + z = 1$ in the first octant 7 Mark

36. Find the volume of the tetrahedron bounded by the three coordinate surfaces

$x = 0, y = 0$ and $z = 0$ and the plane

$$x/a + y/b + z/c = 1. \quad 7 \text{ Mark}$$

37. Find the volume of the tetrahedron bounded by the three coordinate surfaces

$x = 0, y = 0$ and $z = 0$ and the plane

$$x/a + y/b + z/c = 1.$$

7 Mark

38. Find the mass if the tetrahedron bounded by the three coordinate surfaces and the plane $x/a + y/b + z/c = 1$, if its density is given by $\rho(x, y, z) = \rho_0(1 + x/a)$. 7 Mark

39. Evaluate $\int_0^1 \int_0^2 \int_0^2 x^2 yz \, dz \, dy \, dx$. 7 Mark

40. Evaluate $\int_1^3 \int_{\frac{1}{x}}^1 \int_0^{\sqrt{xy}} xyz \, dz \, dy \, dx$ 7 Mark

41. Evaluate $\iiint_V \frac{1}{(x + y + z + 1)^3} dx \, dy \, dz$, where V is the volume bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $x + y + z = 1$. 7 Mark

42. Let G be the wedge in the first octant cut from the cylindrical solid $y^2 + z^2 \leq 1$ by the planes $y = x$ and $x = 0$. Evaluate

$\iiint_G z \, dV$ 7 Mark

43. Find the volume of the region D enclosed by the surfaces $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$. 7 Mark

44. Use a triple integral to find the volume of the solid within the cylinder $x^2 + y^2 = 9$ and between the planes $z = 1$ and $x + z = 5$. 7 Mark

45. Using the idea of triple integral, find the volume of the solid enclosed by the sphere $x^2 + y^2 + z^2 = a^2$. 7 Mark

46. Find the volume of the solid enclosed between the paraboloids

$$z = 5x^2 + 5y^2 \quad \text{and} \quad z = 6 - 7x^2 - y^2 \quad 7 \text{ Mark}$$

47. Let G be the wedge in the first octant cut from the cylindrical solid

$$y^2 + z^2 \leq 1 \quad \text{by the planes} \quad y = x \quad \text{and} \quad x = 0. \quad \text{Evaluate the}$$

integral $\iiint_G z \, dV$ by integrating first with respect to x . 7 Mark

48. Find the volume of the region bounded by the paraboloid

$$z = x^2 + y^2 \quad \text{and the plane} \quad z = 2y. \quad 7 \text{ Mark}$$

49. Find the volume of the tetrahedron bounded by the three coordinate surfaces

$$x = 0, \quad y = 0 \quad \text{and} \quad z = 0 \quad \text{and the plane} \\ x/a + y/b + z/c = 1. \quad 7 \text{ Mark}$$

50. What points satisfy the equations $r = 2, \theta = \frac{\pi}{4}$? 3 Mark

51. Sketch the surface $r = 1 + \cos \theta$ 7 Mark

52. Find the Cartesian coordinates of the point with cylindrical coordinates

$$\left(4, \frac{2\pi}{3}, 5\right). \quad 3 \text{ Mark}$$

53. Find the Cartesian equations of the surface whose equation in cylindrical coordinates is

$$r^2 + 4z^2 = 16. \quad 3 \text{ Mark}$$

54. Find the Cartesian equations of the surface whose equations in cylindrical

coordinates is $r^2 \cos 2\theta = z$. 7 Mark

55. Find a Cartesian equation for the surface $z = r^2$ and identify the surface. 7 Mark

56. Find the cylindrical coordinates of the point with Cartesian coordinates $(-5, -5, 2)$. 3 Mark

57. Find the cylindrical coordinates of the paraboloid whose Cartesian equation is $x^2 + y^2 = 4 - z$ 7 Mark

58. Find the cylindrical coordinates of the cylinder whose Cartesian equation is $x^2 + y^2 = 2x$. 7 Mark

59. Translate $x^2 + y^2 + z^2 = 4z$ into cylindrical co-ordinates. 7 Mark

60. Find the cylindrical co-ordinate of the point given by the Cartesian coordinates $(1, 0, 0)$. 7 Mark

61. Find an equation for the circular cylinder $4x^2 + 4y^2 = 9$ in cylindrical coordinates. 7 Mark

62. Find an equation for the cylinder $x^2 + (y - 3)^2 = 9$ in cylindrical coordinates. 7 Mark

63. Describe the set of points $P(\rho, \theta, \phi)$ whose spherical co-ordinates satisfy the equations $\rho = 1$, $\phi = \pi/3$. 7 Mark

64. Translate the equation $\rho = 6 \cos \phi$ into Cartesian and cylindrical equations. 3 Mark

65. Find a spherical coordinate equation for the sphere

$$x^2 + y^2 + (z - 1)^2 = 1 \quad 7 \text{ Mark}$$

66. Find a spherical coordinate equation for the cone $z = \sqrt{x^2 + y^2}$.

7 Mark

67. Translate the equation $x^2 + y^2 + z^2 = 4z$ into two other forms. 3 Mark

68. Translate $z^2 = r^2$ into Cartesian and spherical equations. 7 Mark

69. Use triple integration in cylindrical coordinates to find the volume and the centroid of the solid G that is bounded above by the hemisphere $z = \sqrt{25 - x^2 - y^2}$, below by the xy -plane, and laterally by the cylinder $x^2 + y^2 = 9$

.7 Mark

70. Use cylindrical coordinates to evaluate

$$\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_0^{9-x^2-y^2} x^2 dz dy dx$$

7 Mark

71. Use the spherical coordinates to find the volume of the solid G bounded above by the sphere $x^2 + y^2 + z^2 = 16$ and below by the

cone $z = \sqrt{x^2 + y^2}$

7 Mark

72. Find the volume of the ice-cream-cone shaped solid D cut from the solid sphere $\rho \leq 1$ by the cone $\phi = \pi/3$. 7 Mark

73. Use spherical coordinates to evaluate

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^{\sqrt{4-x^2-y^2}} z^2 \sqrt{x^2 + y^2 + z^2} dz dy dx$$

Module 4

1. Find first, second, third and n^{th} partial sums of the series $\sum_{n=1}^{\infty} u_n$, where

$$u_n = \frac{1}{2^{n-1}} \quad . \quad 3 \text{ Mark}$$

2. Find the n^{th} partial sum of the series $\sum_{n=1}^{\infty} u_n$, where $u_n = (-1)^n$.3 Mark

3. (a) Test the convergence of the series

$$5 - 4 - 1 + 5 - 4 - 1 + 5 - 4 - 1 + \dots \quad 3\text{Mark}$$

- (b) Show that the series $1 + \frac{1}{2} + \frac{1}{2^2} + \dots$ converges and also find its sum. 3 Mark

4. (a) Show that the series

$$1^2 + 2^2 + 3^2 + \dots + n^2 + \dots \text{ diverges to } +\infty \quad . \quad 3 \text{ Mark}$$

- (b) Show that the series

$$\sum_{m=1}^{\infty} \frac{n+1}{n} = \frac{2}{1} + \frac{3}{2} + \frac{4}{3} + \dots + \frac{n+1}{n} + \dots \text{ diverges.}$$

3 Mark

5. S T. The **geometric series**

$$\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + \dots + ar^{n-1} + \dots$$

converges if $|r| < 1$. 7 Mark

6. Show that the series $\frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \dots$ converges. Find its sum.

7 Mark

7. Show that the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n 5}{4^n} = -\frac{5}{4} + \frac{5}{16} - \frac{5}{64} + \dots$$

has the sum -1 . 7 Mark

You drop a ball from a meter above a flat surface. Each time the ball hits the surface after falling a distance h it rebounds a distance rh , where r is positive but less than 1. Find the total distance the ball travels up and

down. Discuss the particular case with $a = 6$ m and $r = \frac{2}{3}$. 7 Mark

8. Find the rational number represented by the repeating decimal

5.232323...

3 Mark

9. Determine whether the series $\sum_{k=1}^{\infty} 5^{2k} 7^{1-k}$ converges. If so, find the sum. 7 Mark

10. Discuss the convergence of the series

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)} + \dots$$

Also find it sum.

$$2 + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \dots + \frac{2}{3^{n-1}} + \dots$$

7 Mark

11. Find the sum

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots + (-1)^{n-1} \frac{1}{2^{n-1}} + \dots$$

7 Mark

12. Find the sum

$$\frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5} + \dots + \frac{1}{(n+1)(n+2)} + \dots$$

7 Mark

In Example 13 - 16, write out the first few terms of each series to show how the series starts. Then find the sum of the series.

13.
$$\sum_{n=0}^{\infty} \frac{(-1)^n}{4^n}$$

3 Mark

14.
$$\sum_{n=1}^{\infty} \frac{7}{4^n}$$

3 Mark

15.
$$\sum_{n=0}^{\infty} \left(\frac{5}{2^n} + \frac{1}{3^n} \right)$$

3Mark

16.
$$\sum_{n=0}^{\infty} \left(\frac{1}{2^n} + \frac{(-1)^n}{5^n} \right)$$

3 Mark

In each of the geometric series in Exercises 17-18, write out the first few terms of the series to find a and r , and find the sum of the series. Then express the inequality $|r| < 1$ in terms of x and find the values of x for which the inequality holds and the series converges.

17.
$$\sum_{n=0}^{\infty} (-1)^n x^n$$

3 Mark

18.
$$\sum_{n=0}^{\infty} 3 \left(\frac{x-1}{2} \right)^n$$

3 Mark

19. Show that the series $1 - 1 + 1 - 1 + \dots$ is not convergent.

3 Mark

20. Show That , For $a \neq 0$, the **geometric series**

$$\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + \dots + ar^{n-1} + \dots$$

diverges if $|r| \geq 1$. 3 Mark

21. For $a \neq 0$, the **geometric series**

$$\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + \dots + ar^{n-1} + \dots$$

converges if $|r| < 1$ and diverges if $|r| \geq 1$. 7 Markj

22. Determine whether the series $\sum_{k=0}^{\infty} x^k$ converges. If so, find the sum.
3 Mark

23. Find the sum of the series $\sum_{n=1}^{\infty} \frac{3^{n-1} - 1}{6^{n-1}}$. 3 Mark

24. Show that the series $\sum_{n=1}^{\infty} \frac{4}{2^{n-1}}$ converges. 3 Mark

25. Determine whether the series $\sum_{n=1}^{\infty} \frac{7}{n}$ converge or diverge. 3 Mark

26. Determine whether the series $\sum_{n=100}^{\infty} \frac{1}{n}$ converge or diverge. 3 Mark

27. The series $\sum_{n=1}^{\infty} \frac{1}{\sqrt[5]{n}}$ diverges, being a p -series with $p = \frac{1}{5} < 1$. 3 Mark

28. Find the Sum

(a) $\sum_{n=1}^{\infty} \frac{1}{10^n}$ (b) $\sum_{n=1}^{\infty} \frac{n}{n+1}$ (c): $\sum_{n=1}^{\infty} \frac{3}{\sqrt{n}}$ (d): $\sum_{n=1}^{\infty} -\frac{1}{8^n}$

(e) $\sum_{n=2}^{\infty} \frac{\ln n}{n}$ (f): $\sum_{n=1}^{\infty} \frac{2^n}{3^n}$ (g) $\sum_{n=0}^{\infty} \frac{-2}{n+1}$ (h) $\sum_{n=1}^{\infty} \frac{2^n}{n+1}$ (i) $\sum_{n=2}^{\infty} \frac{\sqrt{n}}{\ln n}$

(j) $\sum_{n=1}^{\infty} \frac{1}{(\ln 2)^n}$ (k) $\sum_{n=3}^{\infty} \frac{\left(\frac{1}{n}\right)}{(\ln n)\sqrt{\ln^2 n - 1}}$ (l)

$\sum_{n=1}^{\infty} n \sin \frac{1}{n}$ (m) $\sum_{n=1}^{\infty} \frac{e^n}{1+e^n}$ (n) $\sum_{n=1}^{\infty} \frac{8 \tan^{-1} n}{1+n^2}$

(o) $\sum_{n=1}^{\infty} \operatorname{sech} n$

15* 3 Mark

33. Does the following series converge?

$$8 + \frac{2}{7} + 1 + \frac{1}{3} + \frac{1}{2} + \frac{1}{3!} + \frac{1}{4!} + \dots + \frac{1}{n!} + \dots$$

7 Mark

34. Which of the following series converge, and which diverge?

a)

$$\frac{3}{4} + \frac{5}{9} + \frac{7}{16} + \frac{9}{25} + \dots = \sum_{n=1}^{\infty} \frac{2n+1}{(n+1)^2} = \sum_{n=1}^{\infty} \frac{2n+1}{n^2 + 2n + 1}$$

b)

$$\frac{1}{1} + \frac{1}{3} + \frac{1}{7} + \frac{1}{15} + \dots = \sum_{n=1}^{\infty} \frac{1}{2^n - 1}$$

c)

$$\frac{1+2\ln 2}{9} + \frac{1+3\ln 3}{14} + \frac{1+4\ln 4}{21} + \dots = \sum_{n=1}^{\infty} \frac{1+n\ln n}{n^2 + 5}$$

7 Mark

35. Determine whether the series $\sum_{k=1}^{\infty} \frac{1}{2k^2 + k}$ converge or diverge.

7 Mark

36. Test the convergence of the series

$$\frac{3}{4} + \frac{5}{9} + \frac{7}{16} + \frac{9}{25} + \dots = \sum_{k=1}^{\infty} \frac{2k+1}{(k+1)^2} = \sum_{k=1}^{\infty} \frac{2k+1}{k^2 + 2k + 1}$$

7 Mark

37. Test the convergence of the series

$$\frac{1}{1} + \frac{1}{3} + \frac{1}{7} + \frac{1}{15} + \dots = \sum_{k=1}^{\infty} \frac{1}{2^k - 1}$$

3 Mark

38. Test the convergence of $\sum_{n=1}^{\infty} \frac{\sqrt{n+1} - \sqrt{n-1}}{n^3}$ 3Mark

39. Test the convergence of the series

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{3}{2 \cdot 3 \cdot 4} + \frac{5}{3 \cdot 4 \cdot 5} + \frac{7}{4 \cdot 5 \cdot 6} + \dots$$

3 Mark

40. Test for convergence the series

$$\frac{1}{3} + \frac{\sqrt{2}}{5} + \frac{\sqrt{3}}{7} + \frac{\sqrt{4}}{9} + \dots$$

7 Mark

41. Discuss the convergence of $\sum_1^{\infty} \sqrt{\frac{1+4^n}{1+5^n}}$. 7 Mark

42. Discuss the convergence of the series $\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \right)$. 3 Mark

43. Using Comparison Test, discuss the convergence or divergence of the series

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots$$

3 Mark

44. Test the convergence of

$$\sum_{n=1}^{\infty} \frac{1}{(a+n)^p (b+n)^q}$$

(a, b, p, q being all positive). 7 Mark

45. Discuss the convergence of

$$\frac{1 \cdot 2}{3 \cdot 4 \cdot 5} + \frac{2 \cdot 3}{4 \cdot 5 \cdot 6} + \frac{3 \cdot 4}{5 \cdot 6 \cdot 7} + \dots$$

7 Mark

46. Test for convergence or divergence the series

$$\frac{2}{1^h} + \frac{3}{2^h} + \frac{4}{3^h} + \frac{5}{4^h} + \dots$$

7 Mark

47. Discuss the convergence of the series

$$\sum_{n=1}^{\infty} \frac{\sqrt{n+1} - \sqrt{n}}{n^p}$$

7 Mark

48. Does $\sum_{n=1}^{\infty} \frac{\ln n}{n^{\frac{3}{2}}}$ converge?

7 Mark

49. Test the convergence or divergence of the series

$$\frac{1}{1+3} + \frac{2}{1+3^2} + \frac{3}{1+3^3} + \frac{4}{1+3^4} + \dots$$

7 Mark

49. Test the convergence of the series

$$\left(\frac{1}{2} + \frac{1}{3} \right) + \left(\frac{1}{2^2} + \frac{1}{3^2} \right) + \left(\frac{1}{2^3} + \frac{1}{3^3} \right) + \dots$$

7 Mark

50. Test the convergence of the series $\sum_{n=0}^{\infty} \frac{n^3 + 1}{5^n + 1}$ 7 Mark

51. Comment about the convergence or divergence the series $\sum_{n=1}^{\infty} \frac{n^2}{k^n}$ where k is

Any number

7 Mark

52. : Verify that the ratio test fails in determining the convergence of the series

$$\frac{2}{3 \cdot 4} + \frac{2 \cdot 4}{3 \cdot 5 \cdot 6} + \frac{2 \cdot 4 \cdot 6}{3 \cdot 5 \cdot 7 \cdot 8} + \dots$$

7 Mark

53. Examine the convergence of the series

$$\frac{1}{3} + \frac{1 \cdot 2}{3 \cdot 5} + \frac{1 \cdot 2 \cdot 3}{3 \cdot 5 \cdot 7} + \frac{1 \cdot 2 \cdot 3 \cdot 4}{3 \cdot 5 \cdot 7 \cdot 9} + \dots$$

7 Mark

54. Test the convergence of $\sum_{n=1}^{\infty} \frac{n^n}{n!}$. 3 Mark

55. By means of examples, show that Ratio Test is inconclusive for $\rho = 1$.

3 Mark

56. Test for convergence the series whose k^{th} term is $\frac{k^{k^2}}{(k+1)^{k^2}}$. 7 Mark

57. Discuss the convergence of $\sum \frac{1}{\left(1 + \frac{1}{n}\right)^{n^2}}$ 7 Mark

58. Investigate the behaviour (convergence or divergence) of $\sum u_n$, if

$$u_n = \left(\sqrt[n]{n} - 1 \right)^n$$

7 Mark

59. Which of the following series converges, and which diverges?

a) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$ b) $\sum_{n=1}^{\infty} \frac{2^n}{n^2}$ 7 Mark

60. Test for convergence the series $\sum \left(1 + \frac{1}{\sqrt{n}} \right)^{-n^{\frac{3}{2}}}$ 3 Mark

61. Show that the series $\sum \frac{[(n+1)r]^n}{n^{n+1}}$ is convergent if $r < 1$ and divergent if $r \geq 1$. 7 Mark

62. : Examine the convergence of the series

$$\sum \frac{(n+1)(n+2) \cdots (n+n)}{n^n}$$

3 Mark

63. Discuss the convergence of the series $\sum \left(1 + \frac{1}{n} \right)^{-n^2}$ 3 Mark

64. Let $u_n = \begin{cases} n/2^n & n \text{ odd} \\ 1/2^n & n \text{ even} \end{cases}$

(a) Using the Root Test, show that $\sum u_n$ converges? 3 Mark

- (b) Can the convergence of the series be handled by Integral Test or the Ratio Test? 3 Mark

65. Examine the convergence of the series:

$$\frac{1}{1^p} - \frac{1}{2^p} + \frac{1}{3^p} - \frac{1}{4^p} + \dots \quad (p > 0)$$

3 Mark

66. Test the convergence of

$$\frac{1}{x} - \frac{1}{x+a} + \frac{1}{x+2a} - \frac{1}{x+3a} + \dots$$

x and a being positive quantities.

3 Mark

67. Test the convergence of $\sum_{n=2}^{\infty} \frac{(-1)^n}{\log n}$

3 Mark

68. Discuss the convergence of the series

$$\frac{2}{1^2} + \frac{3}{3^2} + \frac{4}{5^2} + \frac{5}{7^2} + \dots$$

3 Mark

69. Examine the convergence of the series:

$$\frac{x}{1+x} - \frac{x^2}{1+x^2} + \frac{x^3}{1+x^3} - \dots \quad (0 < x < 1)$$

3 Mark

70. Test whether the series

$$1 + \frac{1}{2^2} - \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} - \frac{1}{7^2} - \frac{1}{8^2} + \dots$$

is absolutely convergent or not? Does the series converge?

3 Mark

71. Examine the convergence or divergence of the series

$$\sum_{n=1}^{\infty} (-1)^n \frac{(2n-1)!}{3^n}$$

3 Mark

Module 5.

1. Find the Taylor series generated by $f(x) = \frac{1}{x}$ at $x_0 = 2$. 3 Mark

2. Find the Taylor series and Taylor polynomials generated by $f(x) = \cos x$ at $x = 0$. 7 Mark

3. Find the Taylor series and Taylor polynomials generated by the exponential function $f(x) = e^x$ at $x_0 = 0$. 7 Mark

4. Using Maclaurin's series expand $\log(1+x)$. 3 Mark

$$\frac{1}{1+x}$$

5. Using Maclaurin's series expand $\frac{1}{1+x}$. 7 Mark

6. Using Maclaurin's series find the expansion of $\sin x$. 3 Mark

7. Find the expansion of $(1+x)^m$, using Maclaurin's series. 7 Mark

8. Using Maclaurin's series expand $\tan^{-1} x$. Hence deduce the series

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

7 Mark

8. Find the interval of convergence and radius of convergence of the following power series.

$$(a) \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots + x^n + \dots \quad (b)$$

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$(c) \sum_{n=0}^{\infty} n! x^n = 1 + x + 2!x^2 + 3!x^3 + \dots \quad (d)$$

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

(e)
$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-1}}{2n-1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

7 Mark

9. Find the Fourier series for the 2-periodic function

$$f(x) = \begin{cases} 0, & \text{when } -1 < x < 0 \\ 1, & \text{when } 0 < x < 1 \end{cases}$$

7 Mark

10. Find the Fourier series of the function

$$f(t) = \begin{cases} 0, & \text{when } -2 < t < -1 \\ k, & \text{when } -1 < t < 1 \\ 0, & \text{when } 1 < t < 2 \end{cases}$$

with period 4.

7 Mark

11. (Square wave) Find the Fourier series of f given by

$$f(x) = \begin{cases} -k, & \text{when } -\pi < x < 0 \\ k, & \text{when } 0 < x < \pi \end{cases} \quad \text{and}$$

$$f(x + 2\pi) = f(x)$$

and hence deduce the Madhav-Gregory

series
$$\frac{\pi}{4} = 1 - \frac{1}{3} - \frac{1}{5} - \dots$$

7 Mark

12. Find the Fourier series expansion for x^2 in $-\pi \leq x \leq \pi$.
Hence deduce that

$$(i) \quad 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2} - \dots = \frac{\pi^2}{12}.$$

$$(ii) \quad 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{6}.$$

$$(iii) \quad 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}. \quad 7 \text{ Mark}$$

13. Find the Fourier series of the function

$$f(x) = \begin{cases} x + x^2 & -\pi < x < \pi \\ \pi^2 & \text{when } x = \pm\pi \end{cases}$$

Deduce that $1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}.$ 7 Mark

14. Find a_0 and a_n if

$$x + x^2 = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

in $-\pi < x < \pi$ with period 2π . 7 Mark

15. Find the Fourier series of the function

$$f(x) = \begin{cases} 0 & -\pi < x \leq 0 \\ \frac{\pi x}{4} & 0 < x < \pi \end{cases} \quad \text{and}$$

$$f(x + 2\pi) = f(x). \quad 7 \text{ Mark}$$

16. Find the Fourier series of f defined by

$$f(x) = e^x \quad \text{in } (-\pi, \pi) \quad \text{and } f(x + 2\pi) = f(x). \quad 7 \text{ Mark}$$

17. Find the Fourier series of f given by

$$f(x) = \begin{cases} 1, & \text{when } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ -1, & \text{when } \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases} \quad \text{and}$$
$$f(x) = f(x + 2\pi). \quad 7 \text{ Mark}$$

18. Find the Fourier series of f given by

$$f(x) = \frac{1}{2}(\pi - x) \quad \text{when } 0 < x < 2\pi \quad \text{and}$$
$$f(x) = f(x + 2\pi). \quad 7 \text{ Mark}$$

19. Find the Fourier series of the function

$$f(x) = x \sin x, \quad 0 < x < 2\pi$$

Deduce that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$ 7 Mark

20. Find the Fourier series expansion of

$$f(x) = \begin{cases} \sin x, & 0 < x < \pi \\ 0, & \pi < x < 2\pi. \end{cases}$$

Also evaluate the series $\frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots$ 7 Mark

21. 1 Express the function

$$f(x) = x^2, \text{ when } -1 < x < 1$$

as a Fourier series with period 2. 7 Mark

22. If $f(x)$ is a periodic function with period $2l$, find the Fourier series of $f(x) = x^2$, when $-l < x < l$. 7 Mark

23. Express the function

$$f(x) = x, \text{ when } -1 < x < 1$$

as a Fourier series with period 2. 7 Mark

24. Find the Fourier series expansion for $f(x) = x^2$ in $[-\pi, \pi]$ with $f(x) = f(x + 2\pi) \quad \forall x \in \mathbb{R}$.

Hence deduce that

$$(i) \quad 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2} - \dots = \frac{\pi^2}{12}.$$

$$(ii) \quad 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{6}.$$

$$(iii) \quad 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}. \quad 7 \text{ Mark}$$

25. Find the Fourier series for $f(x) = |x|$ in $[-\pi, \pi]$ with $f(x) = f(x + 2\pi) \quad \forall x \in \mathbb{R}$ and deduce that

$$1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}. \quad 7 \text{ Mark}$$

26. Find the Fourier series of $f(x) = \begin{cases} -x, & -\pi \leq x < 0 \\ x, & 0 \leq x \leq \pi \end{cases}$ and $f(x + 2\pi) = f(x)$. 7 Mark

27. Find the Fourier series for $f(x) = x \sin x$ in $[-\pi, \pi]$ with $f(x) = f(x + 2\pi) \quad \forall x \in \mathbb{R}$. 7 Mark

28. Find the Fourier series of f given by $f(x) = x$, 3 Mark

29. Find the Fourier series of f given by $f(x) = \sinh ax$,

where $-\pi < x < \pi$ and $f(x) = f(x + 2\pi) \quad \forall x \in \mathbb{R}.$ 7 Mark

30. Find the half range expansions of the function

$$f(t) = \begin{cases} \frac{2k}{l}t, & \text{when } 0 \leq t < \frac{l}{2} \\ \frac{2k}{l}(l-t), & \text{when } \frac{l}{2} \leq t < l \end{cases} \quad 7 \text{ Mark}$$

31. Obtain the half range cosine series of

$$f(x) = x, \text{ when } 0 < x < 2. \quad 7 \text{ Mark}$$

32. Find the half range sine series of the function

$$f(x) = \begin{cases} x, & \text{when } 0 < x < 1 \\ 2-x, & \text{when } 1 < x < 2 \end{cases} \quad \text{in } 0 \leq x \leq 2. \quad 7 \text{ Mark}$$

33. Obtain the (i) Fourier sine series and (ii) Fourier cosine series for the function

$$f(x) = x \quad \text{for } x \in [0, \pi] \quad 7 \text{ Mark}$$

34. Obtain the (i) Fourier sine series and (ii) Fourier cosine series for the function

$$f(x) = \pi - x, \quad 0 < x < \pi.$$

where $-\pi < x < \pi$ and

$$f(x) = f(x + 2\pi) \quad \forall x \in \mathbb{R}. \quad 7 \text{ Mark}$$

35. Obtain the Fourier sine series for the function

$$f(x) = c, \quad x \in [0, \pi].$$

Also deduce that
$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

7 Mark

36. Expand $f(x) = \cos x$ in the half range sine-series in $0 \leq x < \pi$.

Example 8 Use Parseval's theorem to show that the sum of the series

$$1 + \frac{1}{2^4} + \frac{1}{3^4} + \dots = \frac{\pi^4}{90}$$

by using the Fourier series expansion of $f(x) = x^2$,
 $-\pi < x < \pi$.

7 Mark